AI-Based Employee Performance Analytics
& Recommendation System

End Semester Project Report

1. Introduction

The "AI-Based Employee Performance Analytics & Recommendation System" is a full-stack web application designed to revolutionize how HR departments and managers evaluate and support their workforce. By leveraging modern web technologies (MERN stack) and Artificial Intelligence (OpenRouter/OpenAI API), the system provides deep, automated insights into employee performance, promotion readiness, and personalized learning paths.

2. Objectives

- Centralized Management: To provide a secure, centralized dashboard for managing employee records, skills, and performance metrics.
- AI-Driven Insights: To eliminate human bias and manual effort by using AI to generate performance reviews, training suggestions, and promotion recommendations based on empirical data.
- Modern User Experience: To deliver a premium, responsive, and visually engaging user interface using a modern design system.

3. Technology Stack

- Frontend: React.js (Vite), React Router, Framer Motion, Recharts, Custom CSS.
- Backend: Node.js, Express.js, JWT for authentication, Bcrypt.js for security.
- Database: MongoDB Atlas (Cloud NoSQL Database), Mongoose ODM.
- AI Integration: OpenRouter API (gpt-3.5-turbo).
- Deployment: Render (Cloud Hosting), GitHub (Version Control).

4. System Architecture

The system follows a standard Client-Server architecture:

1. Client (React): Communicates with the backend via RESTful APIs. Manages local state and global authentication state.
2. Server (Express): Processes requests, validates data, interacts with the MongoDB database, and securely communicates with the external OpenRouter AI API.
3. Authentication Flow: Users register/login to receive a JWT. This token is attached to the Authorization header of all subsequent API requests.

5. Key Features

- Secure Authentication: JWT-based login and registration system protecting the admin dashboard.
- Interactive Dashboard: Visual analytics including Department Distribution (Pie Charts) and Top Performers tables.
- Employee CRUD: Full Create, Read, Update, and Delete functionality for employee profiles.
- Search & Filter: Real-time search functionality to filter employees by department.
- AI Recommendation Engine: Generates a comprehensive 4-part analysis for any employee: Promotion Readiness, Learning Path, Manager Feedback Synthesis, Performance Insight Ranking.

6. Conclusion

The project successfully demonstrates the integration of a traditional CRUD web application with modern Generative AI capabilities. The resulting system is a production-ready, highly secure, and visually premium SaaS application suitable for modern HR teams.